

Notes: Teoh, Welch, and Wong (JF, 1998)

Earnings Management and the Long-Run Market Performance of Initial Public Offerings

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1 Paper Overview

1.1 Research question

The paper asks whether accounting choices around an initial public offering predict subsequent long-run stock performance. The central hypothesis is that IPO issuers can temporarily raise reported earnings through positive discretionary accruals, and that investors partly overvalue these earnings because they do not fully separate cash-flow performance from accrual-based earnings management.

1.2 Core conclusion

IPO issuers with unusually high discretionary current accruals in the IPO fiscal year subsequently underperform issuers with conservative accruals. The return gap is large, economically meaningful, and robust across alternative long-run return benchmarks, holding periods, size partitions, book-to-market partitions, and regression specifications. The paper also shows that conservative issuers are more likely to return to the equity market through seasoned equity offerings, suggesting that financial reporting choices around the IPO are connected to future capital-market access.

1.3 Why the setting is useful

IPOs are an especially informative setting for studying earnings management because:

- information asymmetry is high: many IPO firms have little pre-IPO media coverage, short public operating histories, and limited public financial data;
- entrepreneurs, underwriters, and analysts have incentives to support high reported performance around the offering and shortly afterward;
- current accruals can raise earnings without immediately changing cash flows;
- the later reversal of aggressive accruals gives a natural prediction for long-run stock returns.

2 Data, Timing, and Variable Construction

2.1 Sample

The main sample contains 1,649 domestic U.S. IPOs from 1980 to 1992. Firms must have an offer price of at least \$1, market capitalization of at least \$20 million in 1997 dollars, CRSP post-issue stock return data, and COMPUSTAT data sufficient to calculate accrual measures. Closed-end funds are excluded.

2.2 Event-time convention

The fiscal year containing the IPO is year 0. The paper uses the first public financial statement that includes the IPO year, then starts the return measurement window after a reporting lag. This design is important: the accrual signal is public before the measured abnormal returns begin, so the strategy is implementable rather than a hindsight return sort.

2.3 Earnings-management proxy

The main variable is **discretionary current accruals** (DCA), scaled by lagged assets. Current accruals capture changes in working-capital accounts such as accounts receivable, inventory, other

current assets, accounts payable, taxes payable, and other current liabilities. The paper estimates normal current accruals using a cross-sectional modified Jones model within two-digit SIC industries and treats the residual as discretionary current accruals. High DCA means earnings are high relative to cash flows and relative to what industry fundamentals would predict.

2.4 Return measures

The paper uses multiple long-run performance measures because IPO underperformance is sensitive to benchmark choice. The main specifications include cumulative abnormal returns, buy-and-hold abnormal returns, market-adjusted returns, matched-firm returns, and Fama-French factor regressions. The main effect is not tied to a single benchmark.

3 Empirical Logic

3.1 Prediction

If investors fully understand earnings management, high discretionary accruals should not forecast future abnormal returns. If investors partly extrapolate inflated earnings, then aggressive accrual issuers should be overpriced and should underperform once later statements reveal that earnings momentum is not sustainable.

3.2 Comparison groups

The paper sorts IPOs into quartiles by DCA. The lowest DCA quartile is labeled **conservative**; the highest DCA quartile is labeled **aggressive**. Most tests compare the long-run performance of these two extreme quartiles.

3.3 Why current accruals matter most

The paper emphasizes current accruals rather than long-term accruals because managers have more discretion over short-term working-capital items. For example, revenue recognition, receivables, inventories, and payables can affect reported earnings around the IPO more quickly than depreciation or other long-term accrual items.

4 Main Findings

4.1 Finding 1: IPO-year accruals are unusually high and then reverse

DCA is strongly positive in the IPO year and declines over later years. This pattern is consistent with earnings being shifted into the offering period and then reversing afterward.

4.2 Finding 2: aggressive accrual issuers underperform

The strongest result is that the aggressive DCA quartile earns much lower abnormal returns than the conservative DCA quartile over the three years after the IPO-year financial statement becomes public. The return spread is often in the range of 20% to 30%, depending on the benchmark and return metric.

4.3 Finding 3: the result survives expected-return controls

The underperformance is robust to size, book-to-market, Fama-French factor exposures, calendar-time regressions, value-weighting, equal-weighting, and sample partitions. This is important because IPO returns are known to be sensitive to risk adjustment and benchmark construction.

4.4 Finding 4: the effect is stronger for firms where information problems are plausibly larger

The paper finds strong results in smaller and medium-sized IPOs, where information asymmetry and investor attention constraints are likely more severe. Appendix tests show that the relation is not simply driven by a size or book-to-market artifact.

4.5 Finding 5: conservative issuers return more often to the equity market

Conservative issuers complete more subsequent seasoned equity offerings than aggressive issuers. The interpretation is not mechanical, but it is consistent with conservative reporting being associated with better subsequent market performance and stronger future capital-market access.

5 Incremental Contribution

1. **Accounting explanation for IPO underperformance.** Earlier work documents that IPOs underperform in the long run. This paper provides a specific accounting-based predictor of which IPOs underperform the most.
2. **Cross-sectional predictor rather than only average IPO anomaly.** The paper does not merely show that IPOs underperform; it explains cross-sectional variation in underperformance using issue-year discretionary current accruals.
3. **Accrual decomposition in an IPO setting.** The paper adapts discretionary-accrual methods to young public firms and separates current from long-term accruals, showing that current accruals are the most powerful earnings-management proxy.
4. **Capital-market consequences.** The paper links aggressive reporting not only to stock returns but also to future seasoned equity issuance, connecting financial reporting quality to future financing ability.
5. **Robustness across long-run return methods.** Because long-run event-study results are benchmark-sensitive, the paper's use of many return metrics makes the main conclusion more credible.

6 Interpretation and Cautions

6.1 What the evidence supports

The evidence supports the view that investors overestimate the persistence of earnings for IPO firms with aggressive current accruals. The market appears to price at least part of the accrual component as if it were more permanent than it actually is.

6.2 What the evidence does not prove

The paper does not directly observe managerial intent. High DCA could arise from opportunistic earnings management, managerial optimism, or the timing of the IPO during a period of temporarily high working-capital accruals. The return predictability is nevertheless important because it shows that the accrual signal contains information that the market does not fully impound.

6.3 Why the results matter

For investors, the paper warns that high IPO-year earnings can be misleading if earnings are supported by accruals rather than cash flows. For researchers, it shows that accounting choices can help explain long-run anomalies. For corporate finance, it suggests that disclosure quality and future financing outcomes are linked.

7 Extracted Figures and Tables with Interpretation

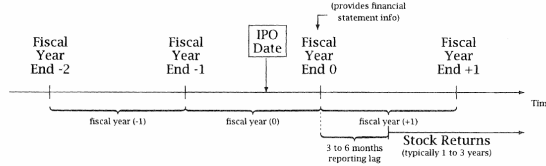


Figure 1. Time line.

Figure 1. Time line

Interpretation. The timeline is central to the research design. The IPO occurs inside fiscal year 0, but returns are measured only after the fiscal-year-0 financial statement is released with a reporting lag. This timing means the DCA signal is observable before the long-run return window begins. The figure helps rule out a purely mechanical look-ahead concern.

Panel B: Time Distribution				
fiscal Year-End	Freq.	%	Cum. Freq.	%
1980	45	2.7	45	2.7
1981	102	6.2	147	8.9
1982	36	2.2	183	11.1
1983	192	11.6	375	22.7
1984	66	4.0	441	26.7
1985	88	5.3	529	32.1
1986	214	13.0	743	45.1
1987	148	9.0	891	54.0
1988	74	4.5	965	58.5
1989	61	3.7	1,026	62.2
1990	76	4.6	1,102	66.8
1991	232	14.1	1,334	80.9
1992	315	19.1	1,649	100.0
Total	1,649	100.0%		

Panel C: Post-Offering Firm Characteristics					
Units	MV (\$m)	BV (\$m)	TA (\$m)	Δ Sales (%)	B/M
mean	199.68	78.68	276.85	80.47	0.400
median	66.23	23.06	39.78	46.30	0.336

Panel D: Immediate Post-Offering Firm Characteristics			
Units	At-IPO Firm Age (years)	Offer Price (\$)	Initial Returns (%)
mean	13.05	12.30	7.12
median	7.00	11.50	1.60

Table I, Panels B-D. IPO timing and firm characteristics

Interpretation. The sample covers hot and cold IPO periods, with especially high volume in years such as 1983, 1986, 1987, 1991, and 1992. The typical IPO is much smaller than the mean IPO, showing strong right skew in firm size. IPO firms also have large sales growth, high valuation ratios, and positive first-day returns, all of which are relevant controls because they can be correlated with both accruals and future returns.

Table I

Sample Characteristics

The sample consists of 1,649 domestic IPO firms going public in the period from 1980–1992 with an offering price of at least \$1 and a market capitalization of \$50 million in 1997 prices (in year 0). The sample firms must also have at least one-month post-issue stock return from CRSP beyond three months after fiscal year 0 (the fiscal year containing the issue date) and sufficient COMPUSTAT data to calculate four components of accruals in fiscal year 0 (see Appendix A for required data items). The distribution of the sample is reported in Panel A by two-digit SIC code, and in Panel B by IPO calendar year. Panel C characteristics of market value, book value of equity, total assets, and ratios Δ Sales (sales change/lagged assets), B/M (book value/market value), and E/P (earnings/price) are measured at the end of fiscal year 0. The change in sales from year –1 to year 0 scaled by assets in year –1. Panel D characteristics are measured at the time of the IPO. Age is the number of years between the founding year or incorporation date (whichever is earlier) and the IPO year.

Panel A: SIC Distribution			
Industry	Two-digit SIC Codes	Freq.	%
Oil and Gas	13, 29	48	2.9
Food Products	20	23	1.4
Apparel and Paper Products	24–27	43	2.6
Chemical Products	28	140	8.5
Manufacturing	30–34	61	3.7
Computer Hardware & Software	35, 73	367	22.3
Electronic Equipment	36	149	9.0
Transportation	37, 39, 40–42, 44, 45	82	5.0
Scientific Instruments	38	130	7.9
Communications	48	41	2.5
Electric and Gas Services	49	16	1.0
Durable Goods	50	45	2.7
Retail	53, 54, 56, 57, 59	135	8.2
Eating and Drinking Establishments	58	40	2.4
Financial Services	61, 62, 64, 65	53	3.2
Recreation Services	70, 75, 79	22	1.3
Health	80	85	5.2
All others	1, 7, 10, 12, 15, 16, 17, 22, 23, 46, 47, 51, 52, 55, 60, 63, 67, 72, 75, 76, 82, 83, 87, 99	169	10.2
Total		1,649	100.0

Table I, Panel A. Sample SIC distribution

Interpretation. The sample spans many industries, but it is not uniform. Computer hardware and software is the largest category, followed by electronic equipment, chemical products, retail, and scientific instruments. This matters because IPO activity is concentrated in growth and technology-oriented sectors; the paper therefore uses industry-adjusted accrual measures and many benchmark controls.

Panel A: Time-Series Distribution of Accruals and Accounting Performance (as a percentage of total assets in the prior year)							
Year	0	1	2	3	4	5	6
Discretionary current accruals (DCA)							
Median	4.01	2.24	0.98	0.14	0.09	−0.51	−0.70
p(sign-rank)	0.000	0.000	0.000	0.089	0.512	0.023	0.695
N	1,649	1,591	1,483	1,372	1,254	943	736
Industry-adjusted net income							
Median	7.82	3.67	1.45	0.23	−0.24	0.18	0.42
p(sign-rank)	0.000	0.000	0.402	0.001	0.000	0.005	0.276
N	1,649	1,592	1,485	1,377	1,256	947	739
Industry-adjusted cash flows							
Median	−4.83	−2.42	−0.99	−0.94	−0.20	0.33	1.18
p(sign-rank)	0.000	0.000	0.000	0.000	0.126	0.252	0.018
N	1,647	1,588	1,484	1,371	1,254	944	738

Panel B: Summary Statistics of Firm Characteristics in Issue Year by DCA Quartile								
Variable	Units (N)	DCA (%)		Std. Dev.	MV* (1997 \$m)	B/M	E/P	NITTA ₁
		Median	Mean			Median	Median	Median
Conservative Q1 (DCA < −5.0%)								
Median	412	−14.93	−24.33	29.54	96.16	31.05	4.38	14.75
Mean	412	−0.12	−0.16	2.42	126.49	38.91	4.44	7.44
Std. Dev.	413	10.10	10.38	4.20	80.02	34.20	4.46	10.62
N	412	39.76	53.92	41.08	65.96	30.60	3.59	17.27
Aggressive Q4 (DCA > 18.5%)								
Median	1,649	4.01	9.95	38.29	90.11	33.60	4.23	11.55

Table II. DCA dynamics and DCA-quartile characteristics

Interpretation. Panel A shows that median DCA is strongly positive in the IPO year and then declines in subsequent years, consistent with accrual reversal. Panel B shows the extreme spread across DCA quartiles: aggressive issuers have very large positive DCA, while conservative issuers have negative DCA. This table motivates the main return tests by demonstrating that the sorting variable has large cross-sectional variation.

Table III
Long-Horizon Mean Abnormal Returns (in percent) by Issue Year DCA Quartiles Using Alternative Benchmarks, Reporting Lags, and Sample Partitions.

The sample consists of 1,649 domestic IPO firms going public in the period from 1980-1992 with an offering price of at least \$1 and a market capitalization of \$20 million in 1997 prices (in year 0). The benchmark for expected returns is the value-weighted market index in all panels except on noted otherwise in Panel A. The holding period is five to forty months after the release of the first post-IPO financial statements in all panels except when noted otherwise in Panel B. Quartile 1 firms are most conservative, quartile 4 most aggressive, in the amount of earnings management the first post-IPO financial statements. The proxy for earnings management (discretionary current accruals, DCA) is described in Table II, and appendix A describes the details of computing it. Cumulative abnormal returns,

$$CAR_{i,t} = \frac{\sum_{t=0}^T (r_{i,t} - m_{i,t})}{N}$$

where $r_{i,t}$ and $m_{i,t}$ are monthly raw and benchmark returns, are statistics computed from the event time-series of firm-average monthly abnormal returns and N is the number of surviving firms in month t . The CAR t -statistics have information on "bad" versus "good" event months. Buy-and-hold abnormal returns,

$$BH_{i,T} = \frac{\sum_{t=0}^T \prod_{s=0}^t (1 + r_{i,s}) - \prod_{s=0}^t (1 + m_{i,s})}{N}$$

when sample firm returns are missing, both $r_{i,t}$ and $m_{i,t}$ are set to zero, are statistics computed from the cross section of multimonth returns net multimonth benchmark returns. Their t -statistics have information on "bad" versus "good" firms. (Due to cross-sectional correlation problems, the t -statistics should not be translated into p values.)

Panel A: Benchmark Variations												
Return	N	Cumulative Abnormal (%)				Buy-and-Hold (%)				Q1	Q2	Q4
		All	Q1	Q2	Q3	All	Q1	Q2	Q3			
w returns	1,649	25.3	31.1	35.4	25.2	9.5	32.2	38.9	40.7	35.1	14.0	
	(t-stat)	(3.77)	(3.58)	(4.39)	(3.60)	(0.99)	(8.45)	(5.53)	(5.94)	(3.93)	(1.88)	
rkt-adj.	1,649	-18.9	-12.0	-7.8	-18.0	-38.2	-15.6	-7.2	-6.1	-11.2	-37.9	
	(t-stat)	(-3.79)	(-1.60)	(-1.24)	(-3.19)	(-4.85)	(-4.12)	(-1.04)	(-0.90)	(-1.26)	(-5.13)	
value-weighted	1,649	-18.9	-12.0	-7.8	-18.0	-38.2	-15.6	-7.2	-6.1	-11.2	-37.9	
	(t-stat)	(-3.79)	(-1.60)	(-1.24)	(-3.19)	(-4.85)	(-4.12)	(-1.04)	(-0.90)	(-1.26)	(-5.13)	

Table III, Panel A. Abnormal returns by benchmark

Interpretation. This is the paper's first major performance table. Across several benchmarks, aggressive DCA IPOs have substantially worse long-run abnormal returns than conservative DCA IPOs. The fact that the sign and economic magnitude survive different benchmark definitions is important because long-run IPO performance is famously sensitive to benchmark choice.

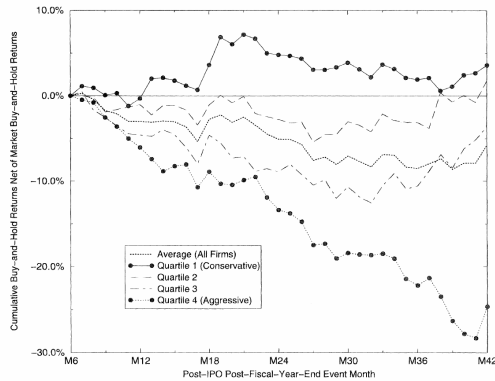


Figure 2. Cumulative buy-and-hold return net of Nasdaq composite index return of DCA quartiles by event month, in percent. The sample is of 1,649 IPOs from the 1980-92 period. (Details are described in Table I.) Firms are divided into quartiles based on how aggressively they manage their earnings. Our measure of earnings management is the firm's discretionary current accruals (DCA), which adjusts for industry, size, and growth. Details are described in Table II and Section I.B.

Figure 2. Cumulative buy-and-hold returns by DCA quartile

Interpretation. The visual pattern is monotonic and economically large. Conservative issuers do relatively well, while aggressive issuers deteriorate steadily after the financial statement becomes public. The widening spread over months M18 to M42 is consistent with investors gradually learning that high accrual-supported earnings are not persistent.

rkt-adj.	1,649	-18.2	-5.2	-4.3	-13.0	-30.6	-7.3	2.4	-0.7	-4.1	-26.8
	(t-stat)	(-4.20)	(-0.71)	(-0.71)	(-1.94)	(-1.85)	(0.35)	(0.10)	(-0.10)	(-1.00)	(-5.69)
Value-Weighted Comp. Ind.	1,649	-12.1	-7.4	-2.6	-12.3	-26.4	-6.3	0.7	2.0	-4.7	-23.2
	(t-stat)	(-2.56)	(-0.99)	(-0.50)	(-2.01)	(-3.55)	(-1.78)	(0.11)	(0.31)	(-0.56)	(-3.34)
Value-Weighted Comp. Ind.	1,649	-8.7	-2.0	-0.7	-27.8	-27.8	0.5	13.5	13.5	-25.0	-25.0
	(t-stat)	(-2.35)	(-0.29)	(-0.18)	(-3.40)	(-3.40)	(0.11)	(1.62)	(1.62)	(-2.39)	(-2.39)

Panel B: Holding Period Variations, Value-Weighted Market-Adjusted Benchmark											
mer	1,649	-21.4	-12.1	-10.6	-20.6	-42.8	-18.3	-5.7	-9.4	-17.0	-41.0
	(t-stat)	(-4.74)	(-1.67)	(-1.84)	(-3.77)	(-5.55)	(-5.13)	(-0.82)	(-1.47)	(-2.20)	(-5.64)
er	1,649	-17.4	-11.6	-5.7	-16.2	-36.5	-14.9	-6.7	-5.5	-10.8	-35.4
	(t-stat)	(-3.39)	(-1.53)	(-0.90)	(-2.75)	(-4.52)	(-3.72)	(-0.90)	(-0.85)	(-1.21)	(-4.24)
ager*	1,649	-22.6	-11.8	-7.9	-23.6	-47.6	-21.1	-7.0	-6.7	-14.4	-56.1
	(t-stat)	(-3.92)	(-1.47)	(-1.01)	(-3.16)	(-4.96)	(-4.14)	(-0.72)	(-0.61)	(-1.22)	(-7.61)

Panel C: Sample Partitions, Value-Weighted Market-Adjusted Benchmark											
firm size in real 1997 dollars	509	-16.9	-21.8	-7.7	-18.6	-28.1	-31.6	-17.3	-28.5	-31.6	-48.7
	(t-stat)	(-2.23)	(-1.64)	(-0.44)	(-1.23)	(-2.00)	(-4.91)	(-1.14)	(-2.29)	(-2.62)	(-5.62)
small firms	888	-25.2	-17.6	-11.4	-21.5	-44.1	-21.9	-4.6	-11.8	-14.0	-49.4
	(t-stat)	(-4.01)	(-1.81)	(-1.30)	(-3.26)	(-4.37)	(-4.10)	(-0.42)	(-1.03)	(-1.13)	(-6.18)
\$20m to \$100m	781	-11.6	-5.8	-5.0	-13.7	-27.4	-8.2	-10.0	-1.8	-7.7	-16.8
	(t-stat)	(-2.43)	(-0.89)	(-0.82)	(-2.22)	(-2.76)	(-1.55)	(-1.21)	(-0.22)	(-0.61)	(-1.14)
book-to-market ratio	817	-23.4	-14.7	-5.1	-14.7	-54.5	-16.2	-15.5	-2.2	-0.9	-40.9
	(t-stat)	(-3.92)	(-2.08)	(-0.57)	(-1.79)	(-4.92)	(-2.76)	(-1.83)	(-0.22)	(-0.06)	(-3.56)
Value*	832	-14.5	-8.9	-9.6	-20.9	-18.9	-14.9	2.9	-8.8	-20.5	-34.2
	(t-stat)	(-2.86)	(-0.87)	(-1.52)	(-3.05)	(-1.97)	(-3.13)	(0.26)	(-0.96)	(-2.30)	(-4.00)
subperiods											
1st half	662	-34.7	-31.7	-20.1	-31.9	-49.6	-40.1	-26.9	-26.7	-42.9	-57.8
	(t-stat)	(-5.01)	(-3.55)	(-2.22)	(-3.41)	(-4.42)	(-3.41)	(-2.93)	(-3.35)	(-3.30)	(-6.41)
2nd half	987	-8.5	-0.0	-1.2	-9.9	-26.9	0.8	5.1	5.0	6.8	-19.0
	(t-stat)	(-1.38)	(-0.00)	(-0.14)	(-1.44)	(-2.54)	(0.15)	(0.54)	(0.53)	(0.64)	(-1.67)

any-year holding periods and IPO firms with less than \$20 million (in 1997 dollars) are not used elsewhere in the paper.

Table III, Panels B-C. Holding periods and partitions

Interpretation. The underperformance is not confined to a single event window. Panel B shows that the return gap appears across different post-statement holding periods. Panel C shows that the effect is visible across firm-size and book-to-market partitions, with especially strong underperformance in the middle-size firms. This reduces concern that the result is only a microcap or valuation effect.

Table IV
Fama-French Three-Factor Monthly Return Time Series Regressions for Size and Book-to-Market Equivalent Conservative and Aggressive (DCA) Portfolios

The sample consists of 1,649 domestic IPO firms going public in the period from 1980 to 1992 with an offering price of at least \$1 and a market capitalization of \$20 million in 1997 prices (in year 0). The table presents time-series regression coefficients from

$$R_{it} - R_{ft} = \alpha + \beta_1(R_{M,t} - R_{ft}) + \beta_2(SMB_{it}) + \beta_3(HML_{it}) + \epsilon_{it}$$

where R_{it} is the value-weighted or equally-weighted return of the aggressive and conservative portfolios in calendar month t , R_{ft} is the return on the value-weighted CRSP index in month t , $R_{M,t}$ is the beginning-of-month three-month T-bill yield in month t , SMB_{it} is the return on small firm minus the return on large firms in month t , HML_{it} is the return on high book-to-market stocks minus the return on low book-to-market stocks in month t (see Fama and French 1992). Firms are considered IPOs from the month in the thirty-month month after the first fiscal year-end. IPO firms in the aggressive and conservative discretionary current accrual size Table II portfolios are selected to minimize size and book-to-market differences in the following way. For minimizing size differences, all IPO firms are first ranked by market capitalization at the end of the fiscal year prior to the IPO. Then, for each category group of IPO firms, the firm with the largest DCA is placed into an aggressive portfolio and the firm with the lowest DCA is placed into a conservative DCA portfolio. The book-to-market adjustment is done similarly by first ranking all IPO firms by their issue year-end book-to-market ratios. The t -statistic in the table is from the intercept in an unintercepted regression of the difference between the conservative and aggressive portfolio returns on the factor returns.

Statistic	Similar MV		Similar BV/MV	
	Conservative	Aggressive	Conservative	Aggressive
Panel A: Value-Weighted Portfolio Returns (in percent)				
α	Cons	-0.002	-0.002	0.028
	p-val	0.939	0.032	0.907
	Cons	0.000	0.000	0.000
	p-val	0.000	0.000	0.000
	Cons	0.000	0.000	0.000
	p-val	0.000	0.000	0.000
	Cons	0.000	0.000	0.000
	p-val	0.000	0.000	0.000
	Cons	0.000	0.000	0.000
	p-val	0.000	0.000	0.000
$\alpha(\text{Aggressive}) - \alpha(\text{Conservative})$		-0.010*		-0.034*
t -statistic		2.180		2.701
Two-tailed p -value		0.032		0.001
Panel B: Equally-Weighted Portfolio Returns (in percent)				
α	Cons	-0.184	-0.853***	-0.013
	p-val	0.000	0.000	0.999
	Cons	1.084***	1.138***	1.001***
	p-val	0.000	0.000	0.000
	Cons	1.228***	1.254***	1.300***
	p-val	0.000	0.000	0.000
	Cons	-0.282***	-0.342***	-0.228***
	p-val	0.000	0.000	0.000
	Cons	0.000	0.000	0.000
	p-val	0.000	0.000	0.000
$\alpha(\text{Aggressive}) - \alpha(\text{Conservative})$		-0.008**		-0.005**
t -statistic		-2.798		-0.087
Two-tailed p -value		0.008		0.002
		156 months: January 1980-December 1995		

***, **, * denote significance at the 1, 5, and 10 percent levels, respectively.

Table IV. Fama-French regressions

Interpretation. The aggressive DCA portfolios have negative factor-adjusted intercepts, while conservative portfolios have much less negative or near-zero intercepts. The difference in alphas remains after controlling for market, size, and book-to-market factors. This table is important because it reframes the return gap as abnormal performance relative to standard asset-pricing factors.

Panel A: Summary Statistics for Sets of 168 Monthly Cross-Sectional Regression Coefficients													
Offset Months	Intercept	log(BV/MV)	log(MV)	if MV < 100	IPO Accruals				All Accruals				
					IDCA	IDLA	INDCA	INDLA	DCA	DLA	NDCA	NDLA	
+6 to +17mo	Coeff	0.184	0.168	0.068	-0.033	-1.569	3.331	-0.244	2.276				
1/82/2/90	(t)	0.155	(6.85)	(2.30)	(-1.13)	(-4.12)	(2.35)	(-0.46)	(1.42)				
+18 to +28mo	Coeff	1.199	0.109	0.035	-0.020	-0.412	5.778	2.239	3.517				
1/83/2/90	(t)	(1.13)	(4.71)	(1.58)	(-0.92)	(-1.96)	(2.55)	(-0.18)	(2.42)				
+30 to +41mo	Coeff	0.798	0.100	0.074	0.012	-2.536	-3.556	-1.481	-1.344				
1/84/12/90	(t)	(0.44)	(5.06)	(2.40)	(0.20)	(-2.38)	(-0.94)	(-0.29)	(0.11)				
+6 to +17mo	Coeff	0.856	0.161	0.069	-0.030	-0.845	2.921	0.135	1.500	-0.774	0.723	-0.471	1.228
1/82/2/90	(t)	(0.29)	(6.89)	(2.32)	(-1.00)	(-2.19)	(1.96)	(0.05)	(0.91)	(-0.99)	(2.65)	(-2.01)	(2.14)
+18 to +28mo	Coeff	1.268	0.107	0.032	-0.021	-0.642	5.194	2.210	3.187	-0.501	0.718	-0.096	0.487
1/83/2/90	(t)	(1.30)	(4.81)	(1.49)	(-0.95)	(-0.79)	(2.01)	(0.09)	(2.15)	(-0.22)	(1.11)	(-0.74)	(0.82)
+30 to +41mo	Coeff	0.869	0.099	0.073	0.011	-2.609	-1.800	-1.531	-1.762	-0.148	0.549	0.024	0.894
1/84/12/90	(t)	(0.61)	(5.25)	(2.32)	(0.12)	(-1.88)	(-1.14)	(-0.19)	(-0.17)	(-1.32)	(1.98)	(-0.12)	(1.61)

Panel B: Univariate Statistics on DCA and IDCA Variables Entering the "+6 to +17mo" Regressions (in percent)													
				Time-Series over 168 Numbers									
Description				Mean	Std. Dev.	Range							
DCA Time-series over 168 cross-sectional averages				1.94	0.52	0.2 to 4.0							
DCA Time-series over 168 cross-sectional standard deviations				31.73	25.18	12.5 to 119.9							
IDCA Time-series over 168 cross-sectional averages				0.37	0.26	-0.1 to 1.0							
IDCA Time-series over 168 cross-sectional standard deviations				7.08	2.67	1.2 to 12.7							
non-zero IDCA's in each month, IDCA time-series over 168 cross-sectional averages				10.96	6.69	-2.8 to 26.3							
non-zero IDCA's in each month, IDCA time-series over 168 cross-sectional standard deviations				36.40	6.88	23.3 to 54.5							

Table V. Fama-MacBeth regressions

Interpretation. The regression evidence shows that issue-year DCA predicts future returns even with controls and with other accrual components included. The negative coefficient on IPO DCA means higher discretionary current accruals are associated with lower subsequent returns. The panel also compares IPO accruals to accruals for all firms, indicating that IPO accruals have distinct predictive power rather than simply repeating a generic accrual anomaly.

Table AI
Calculation of DCA

	Dec. 1990 Year 0	Dec. 1989 Year -1	Difference Δ
Accounts receivable	5.985	2.148	3.837
Inventory	0.091	0.000	0.091
Other current assets	0.721	0.241	0.480
Accounts payable	2.162	1.385	0.777
Taxes payable	0.329	0.455	-0.126
Other current liabilities	0.728	0.920	-0.192
	$\Rightarrow$ IPO Current Accruals (CA):		3.949
Trade receivable	5.985	2.148	3.837
Sales	21.020	11.253	9.767
Total assets	—	20.375	—

Equation (A6) gives IWSI's discretionary accruals as

$$\begin{aligned}
 DCA_{i,t} &\equiv \frac{CA_{i,t}}{TA_{i,t-1}} - NDCA_{i,t} \\
 &= \frac{3.949}{20.375} - \left[\frac{-0.014}{20.375} + 0.121 \left(\frac{9.767 - 3.837}{20.375} \right) \right] = 0.1593.
 \end{aligned}$$

This places IWSI into accruals quartile 3, according to the quartile breakpoints in Table II.

Table AI. Calculation of DCA

Interpretation. This appendix table makes the earnings-management measure concrete. It shows how changes in working-capital accounts are converted into current accruals and then into DCA after subtracting nondiscretionary accruals. The example highlights why receivables and sales growth are central: a large increase in receivables relative to sales can raise reported earnings without the same cash-flow support.

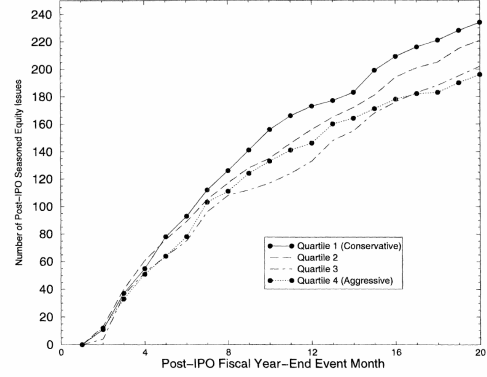


Figure 3. Cumulative seasoned equity offerings (SEOs) post-IPO by event month. The sample are 1,649 IPOs from the 1980-92 period. Details are described in Table I. Firms are divided into quartiles based on how aggressively they manage their earnings. Our measure of earnings management is firms' discretionary current accruals (DCA), which adjusts for industry, size, and growth. Details are described in Table II and Section 1.B. The graph displays the total number of seasoned equity offerings by quarter relative to the first post-IPO financial statements.

Figure 3. Cumulative SEOs after the IPO

Interpretation. Conservative issuers return to the equity market more often than aggressive issuers. The figure suggests a financing consequence of the accounting signal: firms that do not aggressively inflate IPO-year earnings have better subsequent market access, or at least are more likely to conduct SEOs. This complements the return evidence with a corporate-finance outcome.

Table CI
Distribution of Discretionary Current Accruals Means, Three-Year Return Means, and their Correlations in Portfolios Classified by Size and Book-to-Market Quartiles.

The sample here is as described in Table I except that we do not apply the \$20 million size cutoff. Firms are first grouped into four equal-numbered quartiles based on each firm's market value adjusted to December 1997 prices (MV) and into four equal-numbered quartiles based on each firm's book-to-market ratio. Based on these quartile ranks, we construct subgroups ("cells") categorized jointly by both the market value and book-to-market ratios (these subgroups have unequal numbers of firms). For each cell, we then report the mean of our earnings management variable (μ_{DCA}), the three-year buy-and-hold returns net of the respective market index (μ_{BH}), and the within-cell Pearson correlation (ρ_{DCA}) and Spearman correlation (ρ_{sp}).

		B/M Ratio (Low) up to 0.25	B/M Quartile 2 0.25 to 0.36	B/M Quartile 3 0.36 to 0.55	B/M Ratio (High) 0.55 and over	All Firms in Equity Quartile
MV* quartile 1 (small firms) up to \$21.6m	μ_{DCA}	0.159	0.347	0.231	0.183	0.218
	μ_{BH}	-0.546	-0.395	-0.238	-0.236	-0.319
	N	98	88	139	212	537
	ρ_{sp}	-0.143	-0.09	-0.044	-0.129	-0.61
MV* quartile 2 \$21.7m-\$56.3m	μ_{DCA}	-0.037	-0.149	-0.114	-0.167	-0.054
	μ_{BH}	0.357	0.135	0.108	0.108	0.162
	N	106	110	149	173	538
	ρ_{sp}	-0.066	-0.155	0.096	0.002	-0.056
MV* quartile 3 \$58.4m-\$154.6m	μ_{DCA}	-0.172	-0.334	-0.311	-0.036	-0.197
	μ_{BH}	0.089	0.056	0.067	0.027	0.061
	N	104	194	151	89	538
	ρ_{sp}	-0.089	-0.069	-0.055	-0.030	-0.068
MV* quartile 4 (large firms) over \$154.7m	μ_{DCA}	-0.206	-0.127	-0.032	-0.085	-0.118
	μ_{BH}	0.065	0.064	0.068	0.046	0.063
	N	229	146	99	64	538
	ρ_{sp}	-0.051	-0.062	-0.051	-0.157	-0.048
All firms in B/M quartile	μ_{DCA}	-0.096	-0.209	-0.069	-0.241	-0.111
	μ_{BH}	0.145	0.122	0.121	0.117	0.123
	N	537	538	538	538	2151
	ρ_{sp}	-0.086	-0.072	-0.013	-0.086	-0.065
	ρ_{sp}	-0.164	-0.150	-0.118	-0.136	-0.141

Table CI. Size and book-to-market robustness

Interpretation. This robustness table sorts firms jointly by size and book-to-market and reports DCA, three-year returns, and correlations. The negative return-accrual relation appears across many cells, which supports the claim that aggressive accruals forecast underperformance beyond simple size or valuation effects. The result is strongest where information problems are plausibly more severe.

8 Takeaways for Reading and Research Use

- When studying IPO performance, do not treat earnings as a sufficient proxy for performance. Separate cash flows from accruals.
- DCA is an ex ante signal: it is calculated from public financial statements and predicts later returns.
- Current accruals are more informative than total or long-term accruals in this setting because they are easier to manipulate around the offering window.
- Long-run event studies should report multiple benchmarks. This paper's main effect is credible partly because it survives many benchmark choices.
- The results are consistent with investor fixation on reported earnings, but the evidence does not require proving intentional fraud or GAAP violation.

9 Bottom Line

Teoh, Welch, and Wong (1998) provide an accounting-based explanation for cross-sectional variation in IPO long-run underperformance. IPO issuers with high discretionary current accruals around the issue year look strong in reported earnings but subsequently deliver poor stock returns and raise less seasoned equity. The paper's key insight is that the quality of earnings at the IPO matters for both valuation and later financing outcomes.